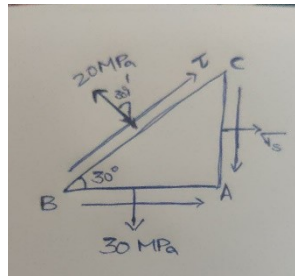


**Q1)** In a certain material under load, plane AB carries a tensile direct stress of 30 MPa and a shear and a shear of 20 MPa, while another plane BC carries a tensile direct stress of 20 MPa and a shear stress. If the planes are inclined to one another at  $30^\circ$  and plane AC at right angles to plane AB carries a direct stress unknown in magnitude and nature, find (a) the value of the shear stress on BC, (b) the magnitude and nature of the direct stress on AC, and (c) the principle stresses.

**Sol)**

Let the stress system be as shown in figure.



(a) To determine the shear stress  $\tau$  on the plane BC, we resolve forces vertically.

$$30 \times AB + 20 \times AC = 20 \times BC \cos 30^\circ + \tau \times BC \sin 30^\circ$$

$$30 \cos 30^\circ + 20 \sin 30^\circ = 20 \cos 30^\circ + \tau \sin 30^\circ$$

$$25.98 + 10 = 17.32 + 0.5 \tau$$

$$\tau = 37.32 \text{ MPa}$$

(b) To find  $\sigma_s$ , we resolve the forces horizontally.

$$20 \times AB + \sigma_s \times AC + \tau \times BC \cos 30^\circ = 20 \times BC \sin 30^\circ$$

$$20 \cos 30^\circ + \sigma_s \sin 30^\circ + \tau \cos 30^\circ = 20 \sin 30^\circ$$

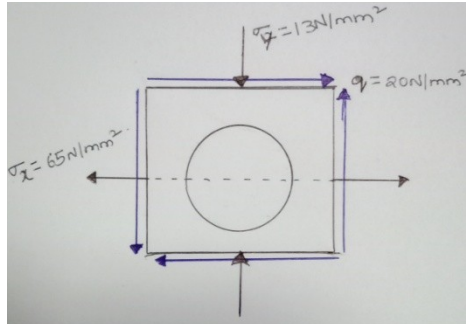
$$17.32 + 0.5 \sigma_s + 32.22 = 10$$

$$\sigma_s = -79.28 \text{ MPa}$$

(c) Two principle stresses are given by:

$$\begin{aligned} \sigma_{1,2} &= \frac{1}{2} (\sigma_x + \sigma_y) \pm \frac{1}{2} \sqrt{(\sigma_x - \sigma_y)^2 + 4\tau^2} \\ &= \frac{1}{2} (-79.28 + 30) \pm \frac{1}{2} \sqrt{(79.28 - 30)^2 + 4 \times 400} \\ &= -24.64 \pm 58.18 \\ &= 33.54 \text{ MPa}, -82.82 \text{ MPa} \end{aligned}$$

**Q2:** A mild steel plate is stressed as shown in a figure. Before stressing, a circle of 300mm diameter is drawn on the plate. Take Poisson's ratio = 0.3; Modulus of elasticity = 200 GN/mm<sup>2</sup>.



Determine:

- The principal stresses developed in MPa respectively.
- The lengths of the major and minor axes, in mm, of the ellipse into which the circle deforms after stressing.

**Solution:**

Given data  $\sigma_x = 65 \text{ MPa}$ ,  $\sigma_y = -13 \text{ MPa}$  and  $\tau_{xy} = 20 \text{ MPa}$

- Principal stresses:

$$\begin{aligned}\sigma_{1,2} &= \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + (\tau_{xy})^2} \\ &= \frac{65 + (-13)}{2} \pm \sqrt{\left(\frac{65 - (-13)}{2}\right)^2 + (20)^2} \\ \sigma_{1,2} &= 70 \text{ MPa}\end{aligned}$$

Strain in the direction of **major** principal stress

$$\begin{aligned}\epsilon_1 &= \frac{\sigma_1}{E} - \mu \frac{\sigma_2}{E} \\ \epsilon_1 &= \frac{70}{200 \times 1000} - \frac{0.3 \times (-18)}{200 \times 1000} \\ &= 3.77 \times 10^{-4}\end{aligned}$$

Change in diameter =  $\epsilon_1$  \* diameter of circle

$$= 3.77 \times 10^{-4} \times 300 \text{ mm} = 0.1131 \text{ mm}.$$

Strain in the direction of **minor** principal stress

$$\begin{aligned}\epsilon_2 &= \frac{\sigma_2}{E} - \mu \frac{\sigma_1}{E} \\ \epsilon_2 &= \frac{-18}{200 \times 1000} - \frac{0.3 \times (70)}{200 \times 1000} \\ &= -1.95 \times 10^{-4}\end{aligned}$$

$$\begin{aligned}\text{Change in diameter} &= \varepsilon_2 * \text{diameter of circle} \\ &= (-1.95 \times 10^{-4}) * 300 \text{mm} = (-0.0585) \text{ mm.}\end{aligned}$$

ii. Therefore,

$$\begin{aligned}\text{Length of the **Major** axis of ellipse} &= 300 + \text{change in diameter} \\ &= 300 + 0.1131 = \mathbf{300.1131 \text{mm}}\end{aligned}$$

$$\begin{aligned}\text{Length of the **Minor** axis of ellipse} &= 300 - \text{change in diameter} \\ &= 300 - 0.0585 = \mathbf{299.94 \text{mm}}\end{aligned}$$

**Q3)** At a point in a loaded components the state of stress is given by  $\sigma_x = 270 \text{ MPa}$ ,  $\sigma_y = 130 \text{ MPa}$

and  $\tau_{xy} = \pm 40 \text{ MPa}$ . Determine

- i) The maximum and minimum principal stresses and the planes on which they act.
- ii) The maximum shearing stress in magnitude and direction.

**Sol:** Given:

State of stress at a point is given by

$$\sigma_x = 270 \text{ MPa}$$

$$\sigma_y = 130 \text{ MPa}$$

$$\tau_{xy} = \pm 40 \text{ MPa}$$

- i) Maximum and minimum Principal Stresses

$$\begin{aligned}\sigma_{1,2} &= \frac{\sigma_x + \sigma_y}{2} \pm \sqrt{\left(\frac{\sigma_x - \sigma_y}{2}\right)^2 + \tau_{xy}^2} \\ &= \frac{270 + 130}{2} \pm \sqrt{\left(\frac{270 - 130}{2}\right)^2 + 40^2} \\ &= 200 \pm 80.622\end{aligned}$$

$$\sigma_1 = 280.62 \text{ MPa}$$

$$\sigma_2 = 119.378 \text{ MPa}$$

For the positions of principal plane

$$\tan 2\theta = \frac{2\tau_{xy}}{\sigma_x - \sigma_y} = \frac{2*40}{270 - 130}$$

$$2\theta = 29.744$$

$$\theta_1 = 14.872^\circ \quad \theta_2 = \theta_1 + 90^\circ = 104.872^\circ$$

ii) Maximum Shearing Stress

$$\tau_{max} = \frac{\sigma_1 - \sigma_2}{2} = \frac{280.62 - 119.378}{2} = 80.62 \text{ MPa}$$

Angular Position of the plane carrying the maximum shearing stress

$$\theta_3 = \theta_1 + 45^\circ$$

$$\theta_4 = \theta_1 + 135^\circ$$

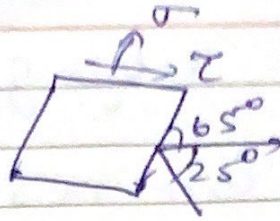
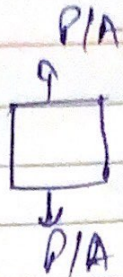
$$\theta_3 = 14.872^\circ + 45^\circ$$

$$\theta_3 = 59.872^\circ$$

$$\theta_4 = 14.872^\circ + 135^\circ$$

$$\theta_4 = 149.872^\circ$$

Q4.



$$\tau_{xy} = 0 \quad \sigma_y = \frac{P}{A} \quad \tau_{xy} = 0$$

$$A = 50 \times 10^{-3} \times 80 \times 10^{-3}$$

↳ area

$$\therefore \sigma = \frac{P}{2A} - \frac{P \cos 130^\circ}{2A}$$

$$P = \frac{\sigma (2A)}{1 - \cos 130^\circ}$$

$$P = \frac{800 \times 10^3 \times 2 \times 50 \times 10^{-3} \times 80 \times 10^{-3}}{1.6427}$$

$$P = 3.9 \times 10^3 \text{ N}$$

$$P = 3.9 \text{ kN}$$



Q5.  $\epsilon_x = \frac{1}{E} (\sigma_x - \nu (\sigma_y + \sigma_z))$

$$\nu = \frac{1}{4}$$

$$\epsilon_x = 37.5 \times 10^{-5}$$

$$\epsilon_y = 0$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

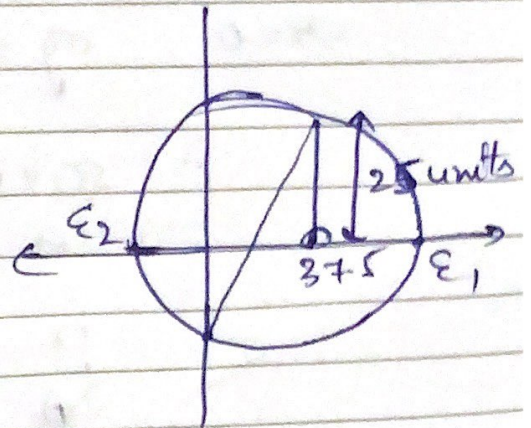
$$G = \frac{E}{2(1+\nu)}$$

• On substituting values

$$G = 80 \times 10^9$$

$$\gamma_{xy} = \frac{\tau_{xy}}{G}$$

$$\gamma_{xy} = 0.5 \times 10^{-3}$$



According to direction  $\Rightarrow \gamma_{xy} = -0.5 \times 10^{-3}$   
 $= -50 \times 10^{-5}$

$$\text{Major axis} = 50 + \epsilon_1 \times 50$$

$$= 50.025 \text{ mm}$$

$$\epsilon_1 = 50 \times 10^{-5}$$

$$\epsilon_2 = -12.5 \times 10^{-5}$$

$$\text{Minor axis} = 50 - \epsilon_1 \times 50$$

$$= 49.99375 \text{ mm}$$